Introduction – types of analytics, applications of predictive analytics, overview of predictive analytics. Setting up the problem - processing steps, business understanding, objectives, data for predictive modeling, columns as measures, target variables, measures of success for prediction

book: Applied Predictive Analytics by Dean Abbott

## DATA SCIENCE

The complete field that combines statistics, programming, domain expertise and data to solve problems.

combines all these methods
to solve real-world problems

### MACHINE LEARNING

Algorithms that learn from data and improve from experience.

**Learns from patterns to**
**Build models**

### PREDICTIVE ANALYTICS

Uses statistical and machine learning techniques to predict future outcomes.

**Predict future events/outcome**

### DATA MINING

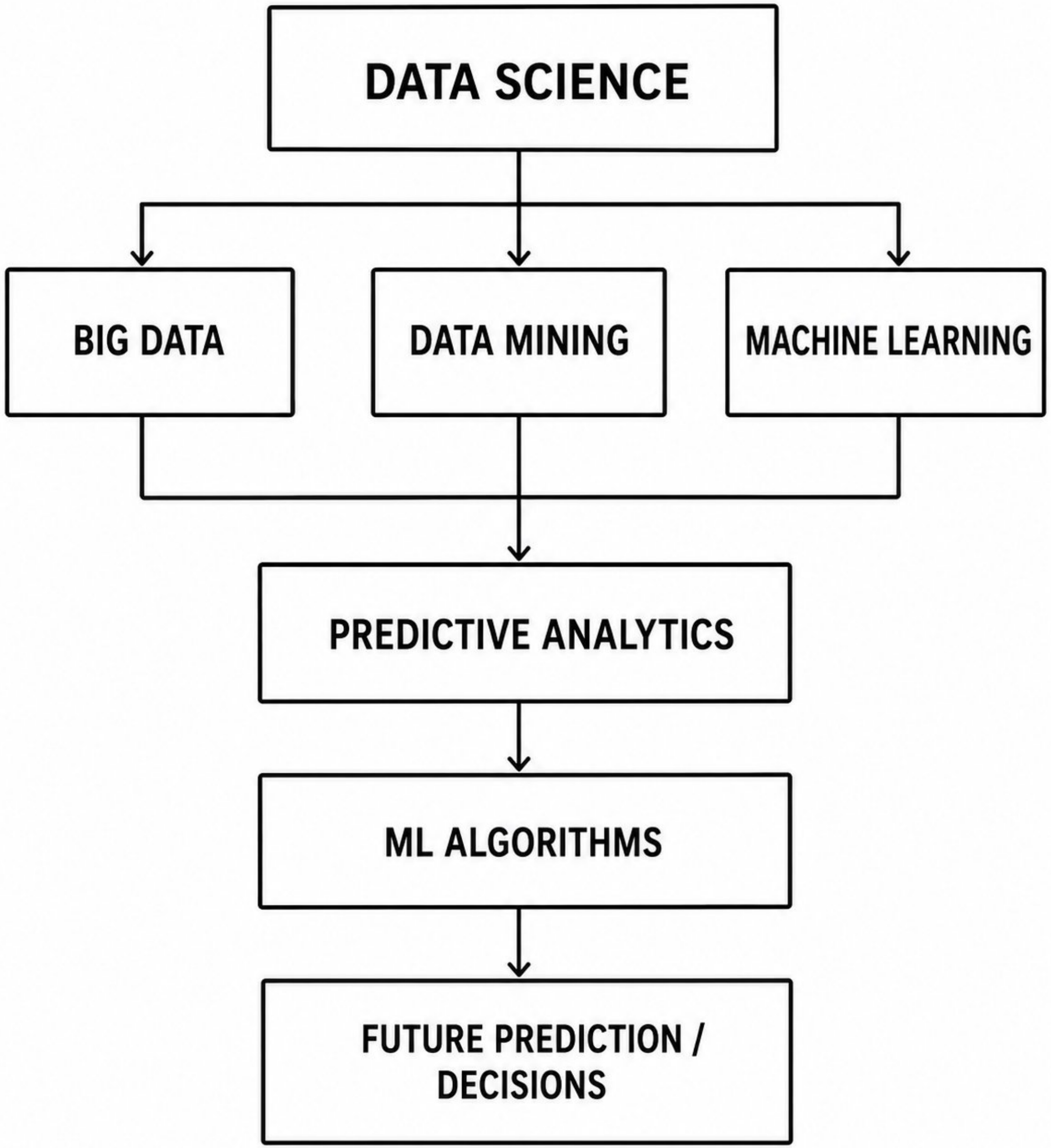
Process of discovering patterns, correlations and useful information from large datasets.

**Find useful/hidden patterns**

### BIG DATA

Large volume, velocity and variety of data from multiple sources.

**Provides huge data and**
**complex data**
**ie** Logs from mobile,
Sensors, Social Media,
Transactions, Images, videos,


DATA SCIENCE
BIG DATA
DATA MINING
MACHINE LEARNING
PREDICTIVE ANALYTICS
ML ALGORITHMS
FUTURE PREDICTION /
DECISIONS

## Uber ride-pricing (how uber uses predictive analytics to predict the best fare before the ride starts.)

Big Data → Data Mining → Machine Learning → Predictive Analytics → Price Decision

1. **Big Data (Collect Huge Data)** : Uber collects millions of records using Ride Data like Pickup location , Drop location , Distance , Traffic condition , Time of day , Weather , Number of drivers nearby , Customer demand, Past ride prices

Eg: (Distance, Time, Traffic, Demand, Price) – (5 km,10 AM,Low,Normal,₹120)
(5 km,6 PM, High, High,₹250)

will become training data

2. **Data Mining (Discover Patterns)**: Uber finds hidden relationships.

Eg: Pattern 1: Office time (6 PM) + Heavy traffic+ More customers ☾ Higher price
Pattern 2: Rainy weather ->More ride requests ->Price increases

3. **Machine Learning model learns**: ML algorithm learns from historical data.

Input: Distance = 10 km , Time = 8 PM, Traffic = Heavy, Demand = High, Drivers = Few

ML models: Regression models/ Decision Trees/ Random Forest/ Neural Networks
Model will learned from the past data trained using features then input given to ML model will predicts expected fare

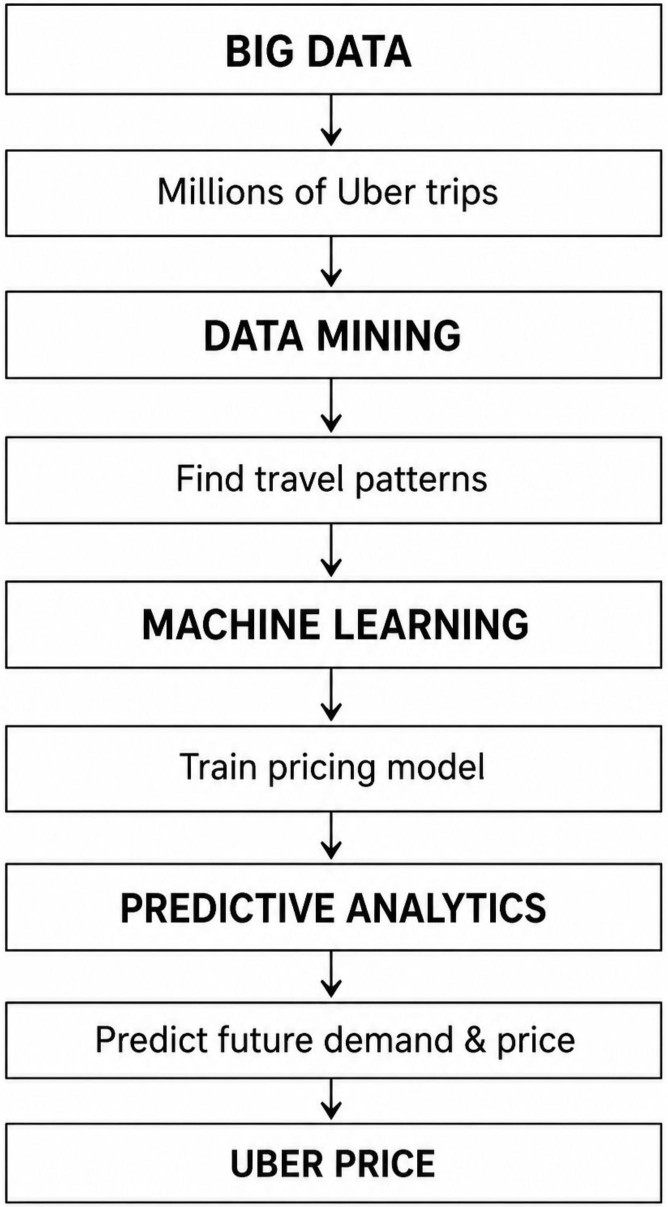
4: **Predictive Analytics**: asks What will the ride price be now?

Current situation:  Customer opens Uber App gives

Location = Airport, Time = 7 PM

Traffic = Heavy, Drivers = Few ☾ Predictive Model ☾ Predicted Price = ₹850

5: **Dynamic Pricing Decision** : Final price will be Base Fare + Distance Cost + Time Cost + Demand Prediction + Traffic Prediction ☾ Final Uber Fare



# Netflix Movie Recommendation

- Big Data : Millions of users, Movie ratings, Watch time, Search history, Likes/dislikes
- Data Mining to finds hidden patterns : User A likes action movies, Similar users watched Movie X
- Machine Learning will Builds recommendation model using: Collaborative filtering, Neural networks
- Predictive analytics will predicts this user has a 90% chance of liking this movie
- Decision will recommend movies on the home page.

# Introduction

- **Analytics** is a term that includes a wide variety of analyses that can be done with data to gain insight. The purpose of analytics is to discover, interpret, and communicate findings to improve the performance ( affect decisions ) of an organization.
- **Predictive analytics** collection of techniques (statistical or ML) that uses historical data to estimate unknown future outcomes so that humans or machines can make better decisions.

- Eg: **Smart Bomb Guidance Problem** : A bomb must reach a target accurately. The ideal path depends on: Current position, Speed, Distance to target, Direction, Control fin movements, earlier scientists used variational calculus to calculate the best path.
- Bomb position +Speed +Target location -> Mathematical Optimization ->Best fin movements ->Hit the target accurately. The simulation worked well. But problem is the calculation was too complex. The small computer inside the bomb could not calculate the best path quickly during flight.
- Complex calculation -> Needs powerful computer ->Bomb computer too small ->Cannot run in real time
- Solution is using Predictive Model Instead of calculating the perfect answer every time Train a model using thousands of previous simulations. The model learns: Situation during flight when (position, speed, distance) given to Predictive Model ->Best control command
- Now the bomb can quickly estimate the best movement. This is called Optimal Path-to-Go guidance

- Eg 2: **Marketing Prediction:** A company wants to know "Will a customer buy/respond? "Historical data: Customer age, Previous purchases, Interest level, Website activity ☾ Predictive Model ☾ Will customer respond?YES / NO . The company can send offers to the right customers.
- What is common between both examples?

Although One predicts human behavior and another One predicts physical movement both follow the same process: Historical Data ->Learn Patterns ->Create Predictive Model ->Predict Unknown Future Value ->Make Better Decision .

**Main Idea is Predictive analytics helps when We have past examples , Future value is unknown, A decision must be made**

# Business intelligence and how it differs from predictive analysis

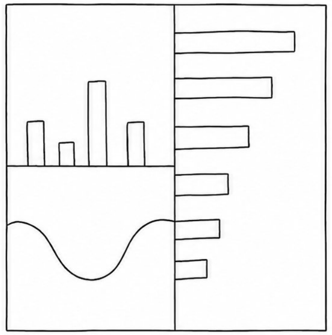
- BI analyzes existing data to understand business performance. It converts raw data into useful information through reports, summaries, and dashboards.BI provides Key Performance Indicators (KPIs) to support decision-making.
- BI mainly answers questions about the past and present (“What happened?”). It helps managers monitor trends, identify problems, and improve business operations. Examples: customer activity like Which product had the highest sales, fraud statistics like How many fraudulent transactions occurred last month?
- BI is retrospective analysis. It analyzes historical data. It explains what happened in the past. Output is usually Reports, Dashboards KPIs
  - Example: How many customers purchased last month? What was last year's sales performance?
- Predictive analytics is prospective analysis. It uses historical data to predict future behavior. It finds patterns in past data and applies them to future situations.
  - Example: Which customer is likely to buy next month? Will a customer respond to an email campaign?

# Types of analytics

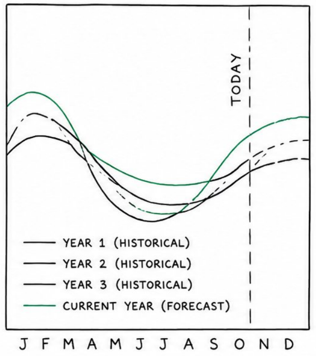
Data analytics techniques are commonly described as part of four distinct categories: descriptive ☾ diagnostic ☾ predictive ☾ prescriptive

1. **Descriptive**: is basic of analytic like cleaning, relating, summarising visualizing your data. questions about what is happening in business
2. **Diagnostic** : getting down to why things are happening. What's causing revenue to decline or to increase? How are things related?
3. **Predictive** what's is going to happen in the future. If you are a business professional and want to make effective business decisions, having an understanding of what might happen in the future can help you ask better questions and make more accurate decisions.
4. **Prescriptive** what is our course of action

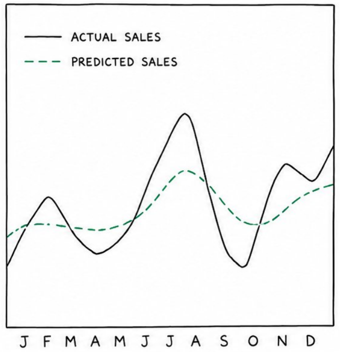
Predictive Analytics Example –
Grocery Store owners



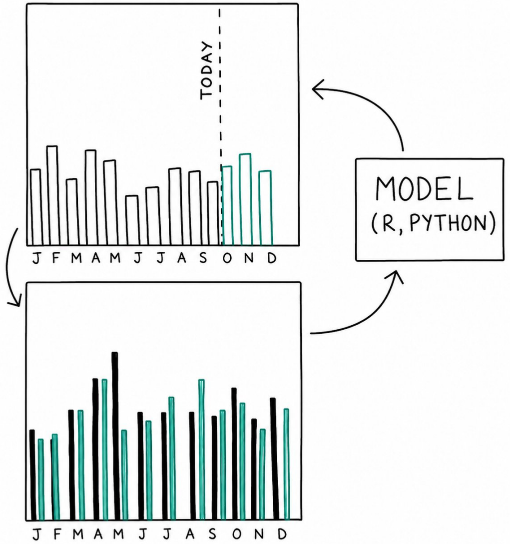
1. Dashboard / BI View




2. Historical Data + Forecast




3. Actual vs Predicted




- A grocery store uses **descriptive analytics** to understand current business performance using dashboards and reports for Sales by department (vegetables, fruits, bakery items) [what is happening], to see Revenue trends and Product performance (product sells more or less) [Why did it happen?]
- The store wants to know or predict **what will happen in the future?** like How many units of each product will each store sell? Which helps in Inventory planning , supply chain management, Profit management
- Building the Predictive Model by Collecting historical sales data from previous years. Analyze past sales patterns and Combine clean data with Mathematical models and Machine learning algorithms
- Using Prediction Results the business can decide whether to increase product stock if demand will rise, plan resources better, Improve profit strategies. [What action should be taken?]
- Model Evaluation : After using the model Compare Predicted Sales (green) vs Actual Sales (black) . If predictions are wrong Identify missing factors, Improve the data/model and Retrain and redeploy. If difference is high add more data and retrain model
- Conclusion: A business person can analyze the prediction chart and make better decisions. ie, if the model predicts that sales will increase, the store can add more products to meet future customer demand. If the prediction shows higher profit than expected, the company can invest in new programs or expand services. Hence predictive analytics helps businesses make smarter and more innovative decisions.

# eg 2: Visiting doctor for annual check up

- DESCRIPTIVE :Your cholesterol level is 215
- DIAGNOSTIC: Your cholesterol level is 215. This is on the high end and likely due to lack of exercise and too much saturated fat in your diet.
- PREDICTIVE: If you maintain your current diet and lifestyle, your cholesterol level will continue to rise and increase your risk of cardiovascular disease.
- PRESCRIPTIVE: Based on your test results, prescribed statins and recommended a new diet to lower your cholesterol and reduce the risk of heart disease.

# Challenges in using Predictive analytics

- Predictive analytics can greatly improve efficiency, decision-making, and return on investment, but many predictive models fail to be successfully implemented in real-world operations

The main reasons for failure include

**1. Obstacles in management**: Here problems are related to people, planning, and business support. ie: Management does not clearly define the problem, No support from decision-makers, Predictions are not used in business decisions.

  - Eg: Suppose an organisation is building a fraud detection model. A fraud detection model can identify 1,000 suspicious transactions every month that further require investigation. For the model to succeed, organizations need proper processes, investigator trust, and management support; otherwise, the predictions may not be used effectively.

**2. Obstacles with data** : Predictive models require data in a structured format with rows and columns (single table or flat file). Data is often stored in multiple transactional databases such as customer, sales, and payment tables. A common key/identifier is needed to combine data from different sources into one table. If proper keys are missing, data cannot be joined and the predictive modeling project may fail. Even after combining data, the information must be Complete , Accurate , Consistent

- Missing or incorrectly stored input/output values make the data unreliable. Poor-quality data leads to inaccurate predictions.
- Eg: data problem in predictive model building for a customer acquisition model. A company wants to predict: Which potential customer (lead) will become an actual customer?

For training, the model needs both examples to learn patterns
(1) People contacted and became customers and (2) People contacted but did not respond

| Lead ID | Age | Income | Campaign | Response |
|---|---|---|---|---|
| 101 | 25 | ₹50,000 | Email offer | Yes |
| 102 | 40 | ₹70,000 | Email offer | No |
| 103 | 30 | ₹60,000 | SMS offer | Yes |

**Problem 1: Data stored in different tables**

Companies may store data separately.

**Marketing Leads Table**

| Lead ID | Campaign | Contacted |
|---|---|---|
| 101 | Email Offer | Yes |
| 102 | Email Offer | Yes |

**Customer Table**

| Customer ID | Name |
|---|---|
| 101 | Ravi |

Issues : If the customer table does not store **which campaign created that customer**, we cannot know:

Which leads responded?

Which leads ignored?

So training data cannot be created.

Problem 2: Updating old data removes history

During campaign:

| Lead | Age | Income | Phone |
|---|---|---|---|
| A | 25 | 40000 | No phone |

Later after becoming customer:

| Customer | Age | Income | Phone |
|---|---|---|---|
| A | 26 | 60000 | 9876543210 |

**The original information is lost.** The model should learn from Data available before prediction is during the campaign, not after.

Problem 2: Data leakage means: Using future information that was not available at prediction time. The company gets phone numbers only after someone becomes a customer. Training data will become

| Lead | Phone Number | Became Customer |
|---|---|---|
| A | Yes | Yes |
| B | No | No |

The model learns: Has phone number -> Will become customer

But this is wrong because the phone number came after conversion.

# Reason for obstacles in data which will lead to wrong prediction

**Different Tables :** Data is stored separately and cannot be linked.

**History Removed:** Old values are overwritten with new ones.

**Data Leakage :** Future information is mistakenly used during training

Predictive models fail when: Old campaign data cannot be reconstructed, Important relationships between tables are missing. Updated data replaces historical values, Future information accidentally enters training data (data leakage).

- **Obstacles with modelling** : first challenges in building predictive models is overfitting. Overfitting occurs when a model becomes too complex and memorizes the training data instead of learning general patterns.
  - An overfitted model gives very good results on training data but performs poorly on new/unseen data. The model's predictions and interpretations become unreliable. If proper testing and validation are not performed during model development, overfitting may only be discovered after deployment. This can cause the predictive model to fail in real-world applications.
- second challenge in predictive modeling is trying to build a model that is too complex or ambitious. Analysts may attempt to create an advanced model without considering Available data, Project requirements , Time limitations
  - If the complex model cannot be completed on time, the project may fail and no model will be deployed. A better approach is to first develop a simple working model that provides useful predictions. The model can be improved later by adding More data , Better features , Advanced algorithms

## Eg: Customer Retention Model Example

Predict which customers may leave the company so actions can be taken to retain them.

- A model developer may collect thousands of possible factors like Customer age ,Purchase history , Website visits ,Number of complaints ,Time spent on app ,Payment history ,Discounts used product ratings etc
- The problem Processing thousands of variables takes more time, Choosing the best combination becomes difficult, The model development may be delayed.
- Better Approach is an experienced analyst already knows some important predictors so used important variables only.

| Variable | Why useful |
|---|---|
| Number of complaints | More complaints → higher chance of leaving |
| Last login date | Inactive users may leave |
| Purchase frequency | Less buying → risk of leaving |
| Customer satisfaction score | Low score → churn risk |

- **Obstacles with deployment :** Predictive models can fail during the deployment stage, even if the model works well during development. The model must be integrated with the organization's operational systems to generate real-time predictions. Models need to be converted into a format or programming language supported by the system.

Eg: SQL , Java , C++ , Other high-level languages

- If the model is not translated correctly, it cannot be used in real-world applications.

Data Preparation Issues During Deployment :

- The same data preparation steps used during model building must be repeated during deployment.
- Data from multiple tables must be Collected , Joined , Converted into the required format
- Complex data processing can delay real-time predictions.

Eg: Website Content Recommendation

- A model predicts: Which content should be shown to a website visitor?
- Required data: Previous browsing history , User preferences , Past interactions

When User visits website ☾ Collect user data ☾ Apply predictive model ☾ Recommend content ☾ Display webpage

Challenge:

- The page must load within seconds.
- Data collection + processing + prediction must happen quickly

# Applications of predictive analytics

**1. Forecasting**

Predicts future values using historical data.

Example: Stock price prediction, sales prediction, weather prediction.

**2. Classification**

Predicts a category/class.

Example: Customer will buy loan → Yes/No, spam detection.

**3. Optimization**

Finds the best possible decision.

Example: Best investment allocation, route planning.

**4. Unsupervised Learning**

Finds hidden patterns without predefined labels.

Example: Customer segmentation, patient grouping.

# Setting up the problem using CRISP-DM process model

- The Cross-Industry Standard Process Model for Data Mining (CRISP-DM) describes the six steps need to be completed in the process of building models.

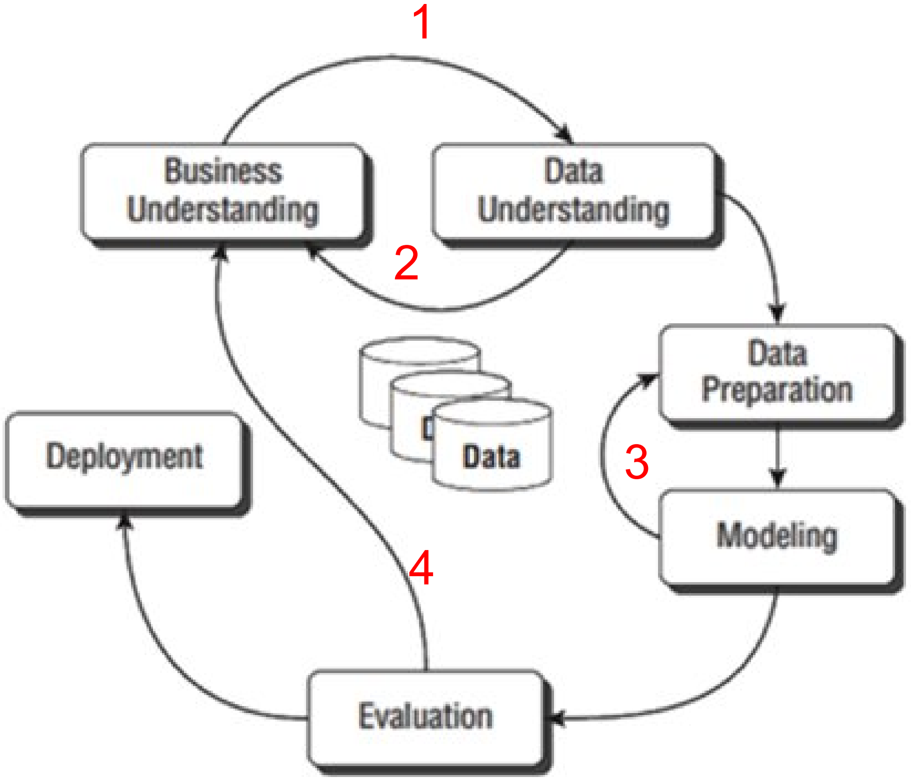



**Table 2-1:** CRISM-DM Sequence

| STAGE | DESCRIPTION |
|---|---|
| Business Understanding | Define the project. |
| Data Understanding | Examine the data; identify problems in the data. |
| Data Preparation | Fix problems in the data; create derived variables. |
| Modeling | Build predictive or descriptive models. |
| Evaluation | Assess models; report on the expected effects of models. |
| Deployment | Plan for use of models. |

- Feedback loops show how the predictive analytics process can be modified and improved during a project.

1. After defining business objectives, the available data is analyzed during the data understanding phase.
2. If the data has problems such as Insufficient quantity and Poor data quality, the business objectives may need to be redefined based on available data. The project may move back to earlier

3. If a developed model gives low accuracy, the data preparation step is revisited. New features or derived variables can be created to improve model performance.
4. Predictive modeling is an iterative process where steps are repeated until better results are achieved.

# Business Understanding

Successful predictive modeling depends on below three important groups of experts (three legged stool) and If any one is missing, the project may fail.

1. Domain Experts Understand the business area and problem. Define the correct business objectives. Ensure the predictive model provides value to the organization.

eg: A banking manager defines the problem how to Predict customers who are likely to fail to repay their loans in the future. Agricultural expert (plant diseases, pest attacks, best harvesting time), Dr (Early disease detection), insurance officer (Predict whether a customer is likely to file an insurance claim, commit insurance fraud, or determine the insurance premium.)

2. Data / Database Experts Identify the data required for building the model. Collect, access, clean, and organize data from different sources. Prepare data in a suitable format for analysis. People who know the databases.

Eg: Collect customer details, transaction history, and loan records.

3. Predictive Modeling Experts develop and test predictive models. Select suitable algorithms and techniques and build models that solve business problems. Machine Learning specialists.

Eg :Use machine learning algorithms to predict loan risk.

Case 1. **If Domain Experts Are Missing** :

- ✓The business problem may not be defined correctly.
- ✓ Modelers may build highly accurate models that do not solve the real business need.
- ✓The model may not provide value to the organization.
- ✓Wrong evaluation measures may be selected, leading to choosing the wrong model.

Eg: Model predicts customer behavior accurately. But prediction does not help improve sales or customer retention then Business does not use the model.

Case 2. **If Data/Database Experts Are Missing**

- ✓Required data may not be identified or accessed properly.
- ✓Analysts may not understand Database structure, Tables and relationships, Meaning of fields
- ✓Problems may occur due to Missing data dictionary, Confusing field names, Lack of data access permissions, Difficulty joining data from multiple tables
- ✓Deployment may fail if the database system cannot support the model requirements.

  Eg: Model needs customer purchase history. Analyst cannot locate or combine the correct database tables.

3. **If Predictive Modeling Experts Are Missing**

✓Business teams may define unrealistic modeling goals.

✓They may expect models that cannot actually be developed.

✓Target variables may be not defined, Poorly defined.

✓Required modeling data structure may not be planned correctly.

✓Important fields needed for prediction may be missing.

Eg: Predict customer leaving. But no proper "customer left/not left" target variable exists.

# Business Objectives

1. **Identify the core business problem** Decide what problem the predictive model should solve. Eg :Predict customers likely to leave. Detect fraudulent transactions.
2. **Measure the business objective** Convert the goal into measurable values. Eg: Reduce customer loss by 20%. Increase sales by 10%.
3. **Identify available data** Check whether required data exists to support the objective. Eg: Customer history, Transactions, Purchase records
4. **Select suitable modeling methods** Decide which predictive techniques can solve the problem. Eg: Classification, Regression, Forecasting
5. **Define model evaluation criteria** Decide how model success will be measured. Eg: Accuracy, Profit improvement, Error reduction
6. **Plan model deployment** Decide how the model will be used in real business operations. Eg: Fraud alerts in banking systems, Product recommendations on websites

Challenges

✓Required data may not be available.

✓Target variable may be difficult to define.

✓Computing resources may be limited.

✓Time and budget constraints may exist

# Defining data for predictive modeling

Predictive modeling requires data in a **two-dimensional format** with Rows , Columns

- Each row represents one unit of analysis (the object/event being studied).
- Data can be loaded from different file formats:

**1. Delimited Flat Files:** Values are separated using delimiters, Common format: CSV files (.csv) , Tab-separated files

**2. Fixed-Width Flat Files:** Each field has a fixed number of characters. No separator is used and the structure must be known before reading data.

**3. Customized Flat Files**

- Special file formats created based on organizational needs.

**4. Binary Files** Software-specific data formats. Like: SPSS files (.sav) , SAS files (.sas7bdat), MATLAB files (.mat)

**Accessing Data from Databases** : Predictive modeling software can directly connect to databases using Native database connections and ODBC drivers, Analysts can write queries to extract required data.

**Advantages of Direct Database Connection**

Avoids saving and loading large files manually.

Data remains stored and managed inside the database.

Reduces problems with maintaining multiple file versions.

Provides analysts more flexibility to select and prepare required data.

# Defining columns as measures

**Columns / Attributes in Data**

- Columns in predictive modeling are also called: Attributes , Variables , Fields , Features ,Descriptors
- Each column describes some information about the unit of analysis.

Eg: Customer Analytics

| Customer ID | Age | Address | Income |
|---|---|---|---|
| 10001 | 35 | Hyderabad | 50000 |

**Rectangular Data Format :** Predictive models require data in **row-column format**.

Every record should have consistent columns

- Same number of columns
- Same order of columns
- Same meaning for each column

**Eg :storing Customers who have visited hotel multiple Visits**

Different customers may have different numbers of visits.

Eg: Customer 1: Visited once 5/2/12

Customer 2: Visited three times 6/9/12, 9/29/12, 10/13/12

## Method 1: Store Visits as Separate Columns

**Table 2-2:** Simple Rectangular Layout of Data

| CUSTOMER ID | DATE OF VISIT 1 | DATE OF VISIT 2 | DATE OF VISIT 3 |
|---|---|---|---|
| 100001 | 5/2/12 | NULL | NULL |
| 100002 | 6/9/12 | 9/29/12 | 10/13/12 |

Problem:
- Creates missing values (NULL).
- Customers with fewer visits have empty fields.
- Some algorithms cannot handle missing values well.

### Method 2: Store Most Recent Visit First

| Customer ID | Recent Visit | Previous Visit |
|---|---|---|
| 100001 | 5/2/12 | NULL |
| 100002 | 10/13/12 | 9/29/12 |

Advantage:
- Recent customer behavior is captured better.
- Useful because latest actions often predict future behavior.

## Method 3: Summarize Visit Information

Instead of storing every visit:
Create useful features.

**Table 2-4:** Summarized Representation of Visits

| CUSTOMER ID | DATE OF FIRST VISIT | DATE OF MOST RECENT VISIT | NUMBER OF VISITS |
|---|---|---|---|
| 100001 | 5/2/12 | 5/2/12 | 1 |
| 100002 | 6/9/12 | 10/13/12 | 3 |

Advantages:

- Removes missing values.
- Maintains rectangular format.
- Easier for predictive models.

# Defining the unit of analysis

- unit analysis defines what each row/record represents, and predictive algorithms assume that every record is an independent observation.
- If records are connected or influenced by each other, the model may learn biased patterns instead of true behavior.
- For eg, Hotel customer data model - built only from employees of one company may learn that company's travel rules rather than general customer travel behavior, making predictions less accurate for other customers.

- It is one of the first decisions made before building a predictive model.

Suppose a hotel wants to build a predictive model to predict:
"Which customer is likely to book a room again?"
Here the unit of analysis = one customer

Each row represents one customer:

| Customer ID | Visits | Purpose | Book Again |
|---|---|---|---|
| 101 | 5 | Vacation | Yes |
| 102 | 1 | Business | No |
| 103 | 3 | Family Trip | Yes |

The model assumes:
Customer 101 behavior is independent
Customer 102 behavior is independent
Customer 103 behavior is independent
It learns general patterns:
- Frequent visits → higher chance of returning
- Long stays → loyal customer

**Problem: Records Are Connected**

Now assume all customers are employees of the **same company**.

| Customer | Company | Visits | Reason |
| --- | --- | --- | --- |
| A | ABC Ltd | 10 | Business meeting |
| B | ABC Ltd | 8 | Training |
| C | ABC Ltd | 12 | Conference |

These customers are connected because:

- Same company books hotels
- Same travel policy
- Same location preference

The model may learn:

ABC company travel rules->Not real individual customer behavior

**After Deployment Problem**

A new normal customer comes:

Family vacation customer-> Model prediction wrong

because the model learned company travel patterns, not general hotel customer patterns.

Each row should represent independent behavior. If records are related, the model may become biased and fail on new data.

## Choosing which Unit of Analysis in Predictive Modeling

- Selecting the correct unit of analysis depends on the **business objective** and how predictions will be used.

**1. Customer as Unit of Analysis**

•Each row represents **one customer**.

Example:

| Customer | Visits | Average Spend | Recent Visit |
|---|---|---|---|
| A | 5 | ₹5000 | Jan |
| B | 2 | ₹2000 | Mar |

•Customer information is summarized into one record.

Features may include: Customer demographics, Number of visits ,Recent visit details, Spending behavior

**Advantages:**

•Gives a complete summary of each customer.

•Each customer record is independent.

**Limitation:**

•Details of individual visits may be lost.

•Extra derived variables are needed to capture trends.

Like ---Increasing spending over visits , Decreasing interest

**2. Visit as Unit of Analysis**

Each row represents **one hotel visit**.

| Customer | Visit | Amount Spent |
|---|---|---|
| A | Visit 1 | ₹2000 |
| A | Visit 2 | ₹5000 |
| A | Visit 3 | ₹7000 |

Example:  A customer with 10 visits will have 10 records.

**Advantages:** Captures detailed information about each visit.

**Limitation:**

•Model may treat repeated visits as separate customers.

•Frequent customers may have more influence on the model.

**Using Derived Variables**

•Derived variables help connect customer history with individual visits.

Eg below details can also be added:

•Total number of previous visits

•Average spending in last 3 visits

•Days since last visit

Important:  Use only past information. Do not include future data (avoid data leakage).

**Choosing the Right Unit**
Depends on the prediction goal:

| **Goal** | **Unit of Analysis** |
| --- | --- |
| Predict customer behavior | Customer |
| Predict next hotel stay | Visit |
| Predict fraud | Transaction |

# Defining target variables in Predictive Modeling

A target variable is the value that a predictive model tries to predict or estimate.

- It is a column in the modeling dataset that represents the business objective.
- Target variables can be: Numeric (continuous) ,Categorical , Binary

**Types of Target Variables**

## Binary Target Variable

- Has only two possible outcomes. (Use: Classification problems)
- Example: Customer Acquisition (1 → Customer acquired , 0 → Customer not acquired)

## Continuous Target Variable

- Predicts a numerical value.
- Example: How much money will each customer spend with us in the future?

| Age | Income | Occupation | Years as Customer | Target (Customer Value |
|---|---|---|---|---|
| 25 | ₹40,000 | Engineer | 2 | **₹50,000** |
| 35 | ₹80,000 | Doctor | 5 | **₹2,00,000** |
| 45 | ₹1,20,000 | Business | 10 | **₹5,00,000** |

Categorical Target Variable Predicts one category among multiple classes.

Eg: Fraud Detection-- Instead of: Fraud = Yes / No

Predict: Fraud Type 1,Fraud Type 2,Fraud Type 3,Fraud Type 4,No Fraud

Advantage: Gives more detailed information for decision-making.

Temporal Considerations for Target Variables : consider the time relationship between input data and the target variable while building a predictive model.

**1. Timeline Arrangement**

A time gap is the time kept between the data used for prediction and the event we want to predict.

This gap allows a business to take action before the event happens.

Eg: Customer Churn Prediction

**Goal:** Predict which customers may stop using a service. (A customer stops using Airtel/Jio SIM. A Netflix subscriber cancels the subscription. A customer closes a bank account.)

Without time gap: Customer behavior data ->Model predicts: "Customer leaves today" -> Too late to take action

- The company cannot stop the customer because the customer has already left.

With time gap: Customer activity history (Jan - March) -> Predictive Model ->Prediction: Customer may leave after 30 days -> (Time gap) Company takes action: - Give discount - Provide special offers - Improve service -> Customer continues using service

Suppose today's date is **30 April**.

### Customer Activity

| Month | Activity |
|---|---|
| January | Uses the service regularly |
| February | Uses the service regularly |
| March | Uses the service less |
| April | Stops using the service |

Now the company builds a model.

**What happens?**

- By the time the model predicts,
- the customer has **already left**.
- The company cannot do anything.

**Input Variable Timeframe -** Defines how much historical information is used for prediction.

Example: Use: Last 6 months purchases , Last 1 year transactions, Recent activities ,Very old data may not be useful.

**Additional Data for Testing**

- Future unseen data is kept aside to test the model.
- Purpose: Check model accuracy, Identify bias, Verify real-world performance.

# measures of success for prediction

- Success Criteria for Classification

✓metrics to assess model accuracy is Percent Correct Classification (PCC). PCC measures overall accuracy without regard to what kind of errors are made; every error has the same weight.

✓Another measure of classification accuracy is the confusion matrix, which enumerates different ways errors are made, like false alarms and false dismissals. PCC and the confusion matrix metrics are good for an entire population in dataset.

For eg, if customers who visit a website are to be served customized content on the site based on their browsing behavior, every visitor will need a model score and a treatment based on that score.

If you are treating a subset of the population, a selected population, sorting the population by model score and acting on only a portion of those entities in the selected group can be accomplished through metrics such as Lift, Gain, ROC, and Area under the Curve (AUC).